

Nafi Ahmed — Portfolio Design System & Architecture Guide

Version: 2.0

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Application: [nafi.dev](#)

1. Executive Summary & Design Philosophy

The portfolio of **Nafi Ahmed** is engineered as a high-performance, immersive single-page web application (SPA) built with **Angular (Standalone Architecture)**, **Tailwind CSS & Vanilla SCSS**, and **Three.js (WebGL)**.

The visual design is grounded in an **editorial luxury tech aesthetic** inspired by modern design studios, characterized by:

- Default Dark Mode** with deep space hues, luminous ambient accents, and ultra-high readability.
- Alternating Cosmic Rhythm**: Odd-numbered sections feature a multi-layered CSS cosmic starfield and shooting stars, while even-numbered sections maintain clean solid surfaces for optical contrast.
- True 3D Interactive WebGL Hero Graphic**: A metallic orbital core sphere surrounded by dynamic tech satellites (Angular, TypeScript, AWS, Flutter, Node.js, Docker).
- Glassmorphism & Optical Depth Bloom**: Subtle semi-transparent surfaces with backdrop blur, glowing borders on hover, and smooth scroll reveals.

2. Color Palette & Design Tokens

The application utilizes a CSS Custom Property architecture defined in `src/styles.scss`, supporting both Dark Mode (default) and Light Mode.

MERMAID

```
graph TD
  A[Root Design System] --> B[Dark Theme (Default)]
  A --> C[Light Theme]
  B --> B1[Base: #090d16]
  B --> B2[Surface: #0e1424]
  B --> B3[Card: #131c2e]
  B --> B4[Primary Accent: #6366f1 Indigo]
  B --> B5[Secondary Accent: #06b6d4 Cyan]
  C --> C1[Base: #f9f7f7]
  C --> C2[Surface: #dbe2ef]
  C --> C3[Card: #ffffff]
  C --> C4[Primary: #3f72af]
  C --> C5[Dark Contrast: #112d4e]
```

2.1 Dark Mode Tokens (Default)

Token	Value	Visual Purpose
--bg-base	#090d16	Deep obsidian base canvas for odd sections and root background
--bg-surface	#0e1424	Midnight navy background for even sections and container panels
--bg-card	#131c2e	Elevated glass card background surface
--bg-card-hover	#19243b	Interactive card hover state
--text-primary	#f8fafc	Pure bright text for headings and primary content
--text-secondary	#94a3b8	Slate text for body paragraphs and descriptions
--text-muted	#64748b	Muted slate for metadata, tags, and microcopy
--border-color	rgba(255, 255, 255, 0.08)	Subtle translucent border for structural division
--gradient-primary	linear-gradient(135deg, #6366f1, #06b6d4)	Signature Indigo-to-Cyan gradient for titles & CTAs
--hero-accent	#38bdf8	Luminous sky blue accent

2.2 Light Mode Tokens

Token	Value	Visual Purpose
--bg-base	#f9f7f7	Soft pearl canvas
--bg-surface	#dbe2ef	Crisp ice-blue surface
--bg-card	#ffffff	Pure white elevated cards
--text-primary	#112d4e	Deep navy for strong contrast typography
--text-secondary	#3e4d61	Charcoal for body copy
--text-muted	#3f72af	Soft oceanic blue for metadata
--gradient-primary	linear-gradient(135deg, #3f72af, #112d4e)	Royal blue gradient

3. Typography & Font Hierarchy

The portfolio employs a curated four-tier font system to establish visual hierarchy and technical authority:

CODE	
Host Grotesk (700-800)	→ Section Titles, Hero Name, Card Headers
Inter (300-600)	→ Body Copy, Paragraphs, UI Text, Navigation
Fira Code (400-500)	→ Tech Badges, Code Snippets, Dates, Period
Azeret Mono (500-700)	→ Status Pills, Badges, Metrics & Numbers

1. **Host Grotesk** (`Host Grotesk` , system-ui, sans-serif): Contemporary geometric display typeface used for `h1` , `h2` , `h3` , and major title anchors.
2. **Inter** (`Inter` , -apple-system, sans-serif): Clean, neutral grotesque font optimized for high legibility across desktop and mobile screens.
3. **Fira Code** (`Fira Code` , monospace): Developer monospace typeface for tech stack chips, timestamps, terminal references, and technical attributes.
4. **Azeret Mono** (`Azeret Mono` , monospace): Stylized industrial monospace used for status capsule pills and metrics.

4. Layout Architecture & Section Rhythm

The homepage is organized into a carefully balanced alternating rhythm that provides visual variety while preserving content hierarchy:

CODE

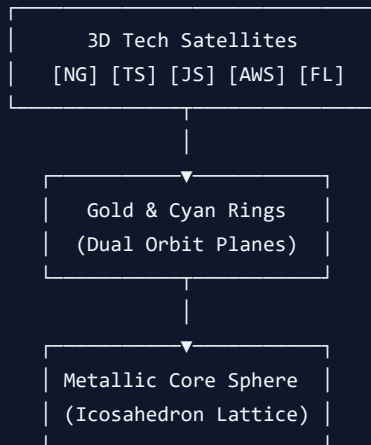
#	Section	Background Type	Visual Characteristic
01	Hero	Starfield (Odd)	3D Sphere, Intro, Live CTAs
02	About	Solid Surface	Bio, Metric Cards, Facts
03	Skills	Starfield (Odd)	Categorized Tech Badges
04	Experience	Solid Surface	Illuminated Timeline Spine
05	Projects	Starfield (Odd)	Keyvo Inset Cards & Links
06	RunMate Club	Solid Surface	Interactive App Showcase
07	Publications	Starfield (Odd)	Academic Research Cards
08	Education	Solid Surface	Degree & Leadership Grid
09	Contact	Starfield (Odd)	Glassmorphism Form & Links
10	Footer	Starfield (Odd)	Brand Signature & Scroll-up

5. Core Visual Elements & Key Components

5.1 Interactive 3D Technology Sphere (Hero Section)

The hero graphic is a custom WebGL Three.js interactive component (`src/app/shared-components/hero-sphere/`):

- Metallic Core:** High-poly sphere (`SphereGeometry(2.5, 48, 48)`) with `MeshStandardMaterial` metalness of `0.92` and subtle counter-rotating icosahedron wireframe lattice.
- Architectural Orbital Rings:**
 - Primary **Gold Ring** (`TorusGeometry(4.1)`) rotated at `68°` with metallic golden specular response.
 - Secondary **Cyan Tech Ring** (`TorusGeometry(5.1)`) rotated at `-45°` with luminous emissive intensity.
- Orbiting 3D Satellites:** 6 tech nodes (**Angular**, **TypeScript**, **Node.js**, **AWS**, **Flutter**, **Docker**) orbiting on distinct elliptical planes with high-resolution canvas monogram sprites.
- Framing & Zero-Clipping Optics:** Configured with `PerspectiveCamera(44°, aspect, 0.1, 100)` at `(0, 0, 18.0)` , guaranteeing full rotation clearance across all display ratios.
- Performance:** Runs outside Angular Zone via `NgZone.runOutsideAngular()` at 60 FPS and automatically pauses rendering when scrolled offscreen via `IntersectionObserver` .



5.2 Cosmic Starfield Engine

Implemented purely in CSS without heavy canvas dependencies, the starfield engine (`src/styles.scss`) renders three continuous parallax layers:

- **Layer 1** (`.stars-sm`): Fine 1px stardust clusters moving on a 90-second linear trajectory with gentle 4-second twinkling.
- **Layer 2** (`.stars-md`): 2px sparkling celestial stars drifting in reverse on a 65-second cycle with scaling micro-pulses.
- **Layer 3** (`.stars-lg`): 3px glowing stars with soft radial aureoles (gold, cyan, and indigo hues).
- **Shooting Stars** (`.shooting-star-track`): High-velocity light streaks rotating at `-35°` with drop-shadow tails firing on 9-second and 14-second intervals.

5.3 Keyvo-Inspired Project Cards (Projects Section)

The project cards (`src/app/pages/projects/`) employ an inset display canvas architecture:

1. **Top Accent Line**: Thin 2px gradient line matching the project's signature brand color that illuminates on hover.
2. **Keyvo Inset Display Canvas**: 110px soft inset preview bay containing an elevated 52px floating icon wrapper with spring rotation.
3. **Floating Quick Actions**: Direct action buttons (GitHub, Google Play, App Store, YouTube, Live Demo) pinned to the top-right corner of the canvas.
4. **Identity & Status Pill**: Role typography accompanied by a status pill capsule (`Office` in indigo or `Personal` in emerald) featuring a live radar dot.
5. **Card Footer**: Clean technology badges paired with an animated "See Details" arrow link.

5.4 Illuminated Timeline Spine (Experience Section)

The experience section (`src/app/pages/experience/`) presents a continuous vertical timeline:

- **Active Role Radar Beacon**: For current employment (`Software Developer @ XORGeek`), the timeline node pulses with an emerald/cyan outer ring (`marker-pulse-ring`).
- **Gradient Spine Line**: Connects timeline nodes seamlessly with a top-to-bottom gradient.
- **Insight Items**: Interactive bullet points with micro-chevron boxes that highlight and translate slightly on hover.
- **Tech Stack Footer**: Dedicated chip list showcasing the specific engineering stack utilized during each tenure.

5.5 Academic Research Showcase (Publications Section)

The publications section (`src/app/pages/publications/`) provides an editorial research layout:

- **Type Ribbon:** Badges classifying papers as `IEEE Conference Paper` or `Journal Article`.
- **Inset Abstract Block:** Highlighted abstract quote with a colored left accent border (`borderLeftColor`).
- **Venue & Indexing Box:** Displays publication citations, volume numbers, and publishing houses clearly.
- **Research Keywords:** Quick-access tags for domains such as Machine Learning, Computer Vision, and Climate Modeling.
- **Dual Action Bar:** Links to the internal deep-dive page and the official DOI/IEEE publication link.

5.6 Academic Credentials & Leadership (Education Section)

The education section (`src/app/pages/education/`) utilizes a responsive two-column grid:

- **Left Column (Academic Background):** Glassmorphism credential cards for B.Sc. Computer Science (AUST), HSC, and SSC with core coursework tags (Data Structures, OOP, Database Systems, Operating Systems, Networks, Software Engineering). **All GPA/CGPA displays are removed for clean professional presentation.**
- **Right Column (Leadership & Extracurriculars):** Vertical leadership timeline showcasing society organizing, innovation club memberships, and workshop facilitation.

5.7 RunMate Club Interactive App Showcase

A custom product section (`src/app/pages/runmate/`) dedicated to the flagship marathon companion platform:

- **Live Device Showcase:** Realistic mobile UI slide carousel displaying real-time GPS tracking, leaderboards, certificates, and event flows.
- **Category Filter Tabs:** Interactive tabs for *Runners*, *Race Organizers*, and *Core Tech*.
- **Tech Stack & Store Badges:** Direct access links to Google Play, Apple App Store, and the Web Admin portal.

6. Micro-Interactions & Animation Guidelines

All animations follow strict timing and easing curves for a cohesive, natural feel:

SCSS

```
// Standard Easing Curves
$ease-spring: cubic-bezier(0.16, 1, 0.3, 1); // Fast out, slow settle (Reveal & Cards)
$ease-bounce: cubic-bezier(0.34, 1.56, 0.64, 1); // Playful snap (Icons & Badges)
$ease-smooth: cubic-bezier(0.4, 0, 0.2, 1); // Standard material transition

// Standard Durations
$duration-fast: 150ms; // Button active states, badges
$duration-medium: 250ms; // Hover transitions, color shifts
$duration-slow: 400ms; // Card hover elevations, theme toggle
$duration-reveal: 700ms; // Scroll reveal blur bloom
```

Optical Depth Bloom (Scroll Reveal)

Elements marked with `.reveal` enter the viewport with zero opacity, a soft `8px` Gaussian blur, and a subtle scale shift (`0.96` to `1.0`), smoothly focusing into view when triggered by the IntersectionObserver.

7. Performance & Accessibility Standards

- **Zero Layout Shift (CLS = 0):** Fixed aspect ratios and explicit SVG bounding boxes prevent content jumping during load.
- **Color Contrast (WCAG AA Compliant):** Text colors ensure minimum 4.5:1 contrast ratios in both light and dark modes.
- **Zero Heavy Image Placeholders:** Clean vector icons (`FontAwesome 6`), CSS radial gradients, and real application assets are utilized exclusively.
- **Dark Mode Default:** Root element is initialized with `data-theme="dark"` to eliminate theme flash during hydration.
- **TypeScript & Linting:** Built with strict typing (`strict: true`), OnPush change detection where applicable, and verified via `npm run lint` with 0 warnings.

8. Summary of File Locations

Feature / System	Key Files
Global Theme & Tokens	src/styles.scss , src/app/services/theme.service.ts
Hero & 3D WebGL Sphere	src/app/pages/hero/ , src/app/shared-components/hero-sphere/
Navigation & Header	src/app/shared-components/navbar/
Project Cards	src/app/pages/projects/ , src/app/pages/projects/project-details/
Experience Timeline	src/app/pages/experience/
Research Publications	src/app/pages/publications/ , src/app/pages/publications/publication-details/
Education & Leadership	src/app/pages/education/
RunMate Product Showcase	src/app/pages/runmate/
Contact & Form	src/app/pages/contact/
Footer Component	src/app/shared-components/footer/
Data Service	src/app/services/portfolio.service.ts